🔷 16.10 Core Structural Components of SEKHEM-TSDD

The SEKHEM-TSDD framework is constructed upon a minimal set of fundamental components that collectively define its functionality as a unified decision–control system.

16.10.1 Core Dynamics (System Evolution Engine)

$$\Psi_{k+1} = \Psi_k + dt\,F(\Psi_k, L_k, P_k, u_k) + \sigma\sqrt{dt}\,\xi_k$$

This equation represents the primary evolution mechanism of the system.

Thus:

the entire system behavior is governed by the structure of F.

16.10.2 Stability Mechanism (Lyapunov Framework)

$$
\frac{d\mathcal{L}}{dt}
\le
- c \|\nabla \mathcal{E}\|^2
+ C\sigma^2
$$

Thus:

stability is guaranteed through energy dissipation under stochastic perturbations.

16.10.3 PAC Mechanism (Triadic Decision Core)

$$D = f(R,C,P), \quad PAC = g(KL)$$

Thus:

decision-making is driven by a triadic structure combining prediction, control, and probabilistic consistency.

16.10.4 Decision Rule

$$
u_k =
\arg\max_{u \in \mathcal{U}_{\text{feasible}}}
$$

$$D_u(k)$$

Thus:

the system selects optimal actions based on evaluated decision scores.

16.10.5 Control and Feedback Injection

$$F \leftarrow F + B(u_k)$$

Thus:

decisions are translated into physical actions that directly affect system dynamics.

16.10.6 Global Closure

$$\mathcal{S}_{k+1} = F_{\text{global}}(\mathcal{S}_k)$$

Thus:

all system components are integrated into a single closed-loop evolution law.

Minimal System Representation

The full SEKHEM-TSDD pipeline can be summarized as:

$$\Psi \rightarrow F \rightarrow PAC \rightarrow D \rightarrow u \rightarrow \Psi$$

with stability ensured through a Lyapunov functional.

Interpretation

The above components constitute the minimal sufficient structure of the SEKHEM-TSDD system.

Thus:

- removing any of these components breaks the system integrity
- additional components (networking, safety, evaluation refinements) enhance performance but are not structurally essential

Closure Statement (Part X Extension)

$$\boxed{
\text{SEKHEM-TSDD can be reduced to a minimal closed-loop decision system while preserving stability, optimality, and adaptivity.}
}$$

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